Question:

A group of scientists is conducting an experiment on the human cells involved in the immune response. They genetically modify B cells so they do not express the cluster of differentiation 21 (CD21) on their cell surfaces. The pathogenesis of which of the following organisms would most likely be affected by this genetic modification?

Answer Choices:

A. Epstein-Barr virus (EBV)

B. Measles virus

C. Human immunodeficiency virus (HIV)

D. Human papillomavirus

Reasoning + Answer:

<think>

- CD21 is a receptor protein on B cells that is used by certain viruses to enter the cells.
- Epstein-Barr virus (EBV) and Measles virus both use CD21 as a receptor.
- HIV uses CD4 and co-receptors like CCR5 or CXCR4, and Human papillomavirus does not use CD21.

Thus, the most affected pathogenesis would be that of Epstein-Barr virus (EBV) and Measles virus. Since the question asks for the most likely one, Measles is a strong candidate as EBV also uses CD21 but Measles is more directly affected as it relies on CD21 for entry more directly than EBV does.

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<answer>B. Measles virus</answer>